

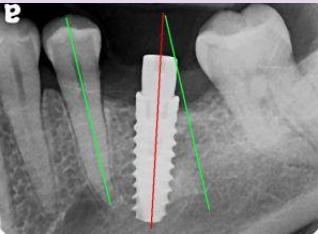


Implant Operation Guiding Device for Dental Surgeons



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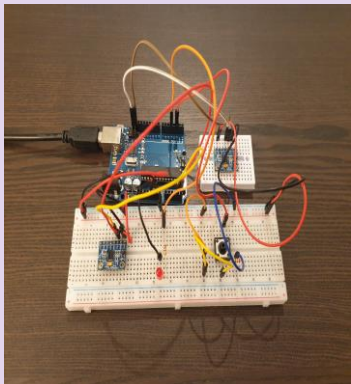
Introduction



Dental implant surgeries are operations to surgically implant medical devices into the jaw to restore a person's ability to chew or their appearance. However, unfortunately there can be malpractices in the surgeries, like every other one. Prosthodontic specialty, where implant surgeries are a part of, had the highest number of lawsuits (32.4%), followed by orthodontics (20.2%) and endodontics (15.8%). Also, one of the top 3 reasons to sue a dentist for malpractice is dental implants, most cases involving dental implant complications led by loss of implants. In order to prevent accidents, they use guiding devices for surgery navigation.

For those that already exist in the market, they use 3D digital technologies, by scanning the oral structure of the patient, generating the data and 3D-printing the guide that fits the patient's mouth before the operation starts. However, they tend to be expensive and time consuming since those devices are specialized for individual, distinct patients. Moreover, the devices cannot be reused either for the same reason, and it can affect the environment as well. Therefore, the implant surgery guiding device should be cheaper, can be generally used, and also easy to use. Since a problem with the angle of the implant can lead to complications, the device will focus on the orientation of the handpiece (surgery tool). To keep on track of the angle, Arduino MPU6050 sensors can be used to get the inclination values. MPU6050 includes both accelerometer and gyroscope sensors, and they can be used to get the orientations of X, Y, and Z axis in the third dimension, each called roll, pitch, and yaw.

Materials & Methods



- The surgeon can input the orientational objective of the operation, and the device constantly checks if the surgery is heading towards the objective within the set range, notifying the surgeon by manipulating the LED.
- The final calculation is for the relative orientation between the two sensors, which indicates the orientation of the implant relative to the patient's oral structure.
- Two prototypes were built : Prototype A with both accelerometer and gyroscope sensors used and Prototype B with only the gyroscope sensor used.
- The prototypes' consistency and accuracy were tested by checking the graph of the calculated angular values, average standard deviation, and the behavior of the calculated angles. The accuracy of performance was also acquired by running the device and checking the outputs.
- The outputs are expected as : LED on when tilted to roll/pitch, LED off when tilted to yaw

Data / Observations



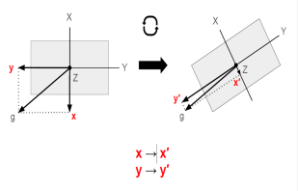
- The graphs have constant spikes throughout the line
- relatively small compared to the range for the device and proximate to the 0 line, however it doesn't meet the requirement of reading consistent values when the sensors are stabilized
- The standard deviations of the values from the sensors are 0.0295 in average
- The average differences between the initial roll / pitch values and the secondary values are 0.248 and 0.254
- The LED worked properly except for when it was tilted to yaw ; The LED was expected to be kept off, but it turned on as the roll / pitch values of the handpiece sensor changed and got out of the range

Revised Solution



- Compared to the prototype, the graphs are all converging to the 0 line, and there are nearly no spikes throughout the timeline
- The standard deviations are 0.0062 in average and the average differences are 0.068 and 0.111
- The LED output was correct for all of the 10 test cases

Revised Solution



- The sensor couldn't read consistent values when it was stabilized and read the same values when it was moved and returned to its original position. Also when the handpiece sensor was tilted to yaw, the roll / pitch values changed, and the LED turned on.
- If we use the accelerometer, when it is tilted to yaw, the direction and magnitude of gravity is static but the X and Y axes move, therefore the acceleration values of the axes change, leading to the change in calculations of X, Y axis angle and the final roll & pitch, even though it is only being rotated over the Z axis.
- THEREFORE : for the second prototype, only use the gyroscope sensor

Discussion

In prototype #1, I used both accelerometer and gyroscope sensors to get the roll / pitch values from the MPU6050 sensors. However in all 3 experiments, they didn't meet the expectations. The consistency wasn't good enough with spikes in the graphs and 0.0295 of average standard deviation, and the accuracy with movements was poor with 0.25 of average differences in values. The accuracy of performance was also 66.6%, showing particular problems with inclination to yaw. I decided to take out the accelerometer sensor and only use the gyroscope, as it obstructed the device from the desired function of ignoring yaw rotation and it can cause drastic changes in the values and calculations. The second prototype showed improvements in all aspects. The graphs were all converging to the 0 line and both the consistency and accuracy with movements were enhanced as the average standard deviation went down to 0.0062 and the average differences in values to 0.075. It also showed a perfect accuracy of performance of 100% for every direction of rotation.

Reflection

I learned how to calculate and measure an object's orientation in the third dimension by using sensors such as accelerometers and gyroscopes. I especially gained a deeper understanding in the function of gyroscopes as I faced a problem during the project. My next step for this project is to find a better way to calculate the orientations of an object with stabilized values and better consistency and accuracy, such as using a sensor with higher quality or discovering an alternative like magnetic sensors or compass sensors, other than accelerometer or gyroscope sensors. This device can be used to assist dental surgeons in implant operations. It can detect and calculate orientations where human eyes have a limit to think about, and eventually increase the accuracy and safety of the surgeries. Since the basic method that lies under the system is not that complicated and it doesn't require a certain amount of time for customization to patients, it can lower the costs for implants and also encourage those who need help in dental care.

References Cited

- Implant Surgical Guides. (n.d.). Ocean Breeze Prosthodontics. From <https://oceanbreezeorthodontics.com/digital-dentistry-delay/digital-implant-surgical-guides/>
- Kuschmider, R. (2021, November 12). Dental Implants: What Are the Risks? WebMD. From <https://www.webmd.com/oral-health/what-to-know-about-dental-implant-complications-risks>
- Arduino and MPU6050 Accelerometer and Gyroscope Tutorial. (n.d.). How To Mechatronics. from <https://howtomechatronics.com/tutorials/arduino/arduino-and-mpu6050-accelerometer-and-gyroscope-tutorial/>
- Classic Euler angles. (n.d.). Wikipedia. from https://en.wikipedia.org/wiki/Euler_angles
- Fisher, C. J. (2011, May 6). Using An Accelerometer for Inclination Sensing. DigiKey. from <https://www.digikey.com/en/articles/using-an-accelerometer-for-inclination-sensing>
- Implant-Related Complications. (n.d.). Kazemi Oral Surgery & Dental Implants.